Local Semantic Map-Based Navigation System for Quadruped Inspection Robot in Complex Terrains

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***Abstract*—Traditional navigation methods provide a convenient and effective approach for quadruped robot navigation. However, the lack of terrain height awareness often leads to misidentifying height variations as obstacles, limiting their effectiveness in inspection tasks. This paper proposes a navigation system for quadruped inspection robots based on local semantic map, utilizing a convolutional neural network-based semantic segmentation algorithm to obtain real-time environmental information. By constructing a multi-level global costmap and a local semantic map through a hierarchical costmap framework, the proposed method enables robust navigation in complex terrains and dynamic obstacle environments. This article was validated in simulated and actual environments including slopes, steps, etc., with a success rate of 85.7%. This approach enhances the adaptability of quadruped robots, addressing navigation challenges while improving inspection efficiency and safety.**

***Index Terms*—Quadruped Inspection Robot, Local Semantic Map, Autonomous Navigation**

I. Introduction

In recent years, quadruped robots have gained popularity in inspection tasks due to their terrain adaptability. However, most navigation methods are purely geometric [1]–[3]. While they perform well in structured or known environments, such as indoor [4] or underground settings [5], they struggle in unstructured outdoor environments [6]. The lack of terrain semantic information often leads to misidentifying varying terrain heights as obstacles, limiting their effectiveness in inspection tasks. Thus, path planning in complex environments remains a key challenge. Currently, quadruped robot navigation relies on laser radar or visual information, with semantic evaluation used to correctly interpret obstacles like steps or stairs.

Terrain Perception Based on LiDAR Technology. Path planning research has long focused on laser navigation. Yang et al. [7] propose a LiDAR-based mapping approach using OctoMap and depth camera matching, though it suffers from high memory usage and lacks semantic information. Gargoum et al. [8] design a spline-based method to extract road cross-sections from LiDAR data for slope classification, but its applicability is limited. In [9], Hu et al. apply polynomial fitting with RANSAC for detecting road edges, but it struggles with curved roads. Du et al. [10] design 2D grid maps for collision-free navigation but fail with elevation differences. Sun et al. [11] create 2.5D elevation maps for multi-terrain navigation, but memory intensity and a focus on static obstacles hinder real-time use. These methods perform well on flat terrain but often misclassify elevated terrain as obstacles, limiting navigation capabilities.

Terrain Perception Based on Vision Technology. Vision sensors are key for quadruped robot navigation. Zhang et al. [12] design a SLAM system using depth camera-based optical flow for 2D grid maps but lack semantic data. Milburn et al. [13] use Mask R-CNN for localization but face issues with dynamic obstacles and limited view. Deng et al. [14] apply CNN-based semantic segmentation to identify paths, but struggle with dynamic obstacles. Fu et al. [15] combine proprioception and vision for path planning, though lack of semantic support reduces safety. Guzzi et al. [16] design local elevation maps for rugged terrains but fail to adapt in dynamic environments. These methods work well in static scenarios but are less effective in dynamic conditions.

Navigation Based on Semantic Information. Using semantic data for robotic navigation is a growing focus. Sun et al. [17] and Goel et al. [18] predict semantic maps for navigation, but their approaches are limited to flat indoor terrains. Chaplot et al. [19] combine semantic segmentation with target-oriented exploration for indoor navigation, but dynamic performance is not well studied. Hong et al. [20] apply contrastive learning to enhance navigation but with limited scope. Guan et al. [21] use visual and inertial data for terrain classification but validate only on tracked robots. While most methods generate global semantic maps, they are memory-intensive and suited mainly for indoor environments, limiting their use in complex, dynamic situations.

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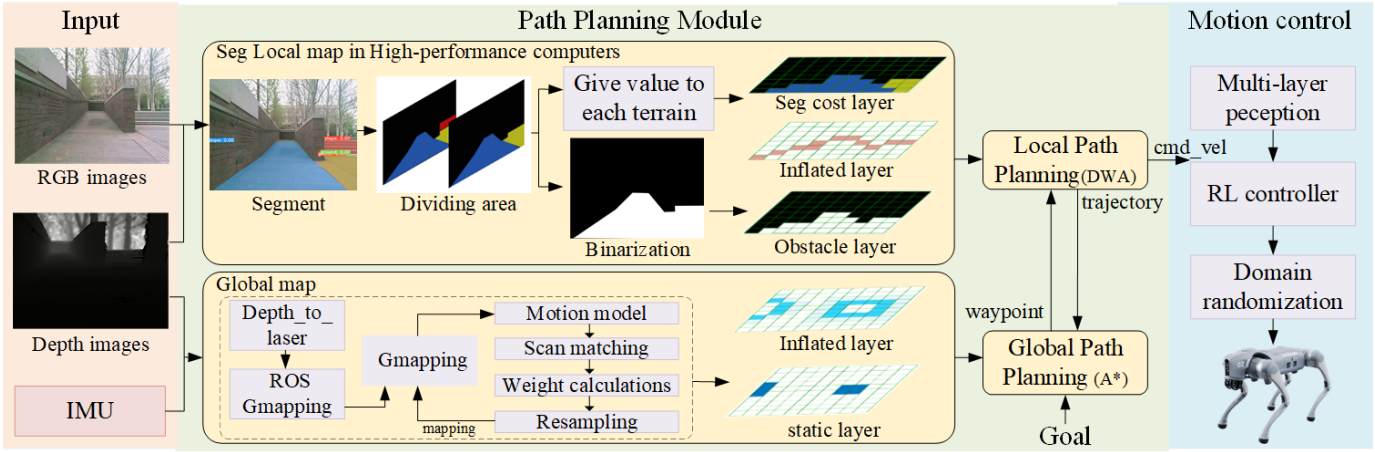


Fig. 1: Quadruped Robot Navigation System.

This paper extends traditional cost maps by introducing multiple cost layers, including a semantic cost layer, to construct both global and local semantic maps. A navigation method based on the local semantic map is then proposed. The contributions of this paper are:

- Semantic segmentation is used to extract terrain information, which is mapped to create obstacle and semantic cost layers, forming a local semantic map with multiple cost layers.
- A path planning method using the local semantic map for local navigation is proposed, enhancing traditional point-to-point navigation.
- Local goal points in global path planning are matched with semantic cost values, and a walking strategy is applied to calculate the robot's velocity in complex terrains for safe navigation.

The paper is structured as follows: Section 2 presents the system framework and local semantic map generation method. Section 3 covers path planning, and motion control. Section 4 validates the method through experiments. Section 5 concludes the paper.

II. Local Semantic Map Generation

To achieve safe navigation in complex terrains, a quadruped robot navigation method based on a local semantic map is proposed, as shown in Fig. 1. The whole framework can be divided into three major components. First component is map construction, in which the local semantic map generation part is discussed in this section. The other two components of the system, path planning and motion control, will be explained in Sec 3.

The semantic segmentation of RGB images is used to identify traversable regions, and a deep data generation obstacle layer and a semantic cost layer are added for map construction. Ultimately, the multi-layer information is integrated to construct the local semantic map.

Based on the idea of hierarchical cost maps [22], this paper introduces semantic information to improve the quadruped robot's adaptability in complex terrains and changing environments. The local cost map contains the obstacle layer obtained from semantic segmentation, the semantic cost layer with different terrain semantic information, and the expansion layer.

A. Obstacle Layer

This paper uses the SparseInst algorithm, which performs excellently in outdoor terrain segmentation tasks, for semantic segmentation to identify traversable terrain and obstacles, thereby constructing the obstacle layer. The specific methods include semantic segmentation, contour extraction, and coordinate mapping.

1) Semantic segmentation:

a) Dataset: The dataset includes images from various terrains (flat land, slopes, grass) with fixed obstacles and dynamic pedestrians under different lighting and weather conditions. It contains over 2000 training images and 400 validation images (640 × 480 resolution), covering all terrain types.

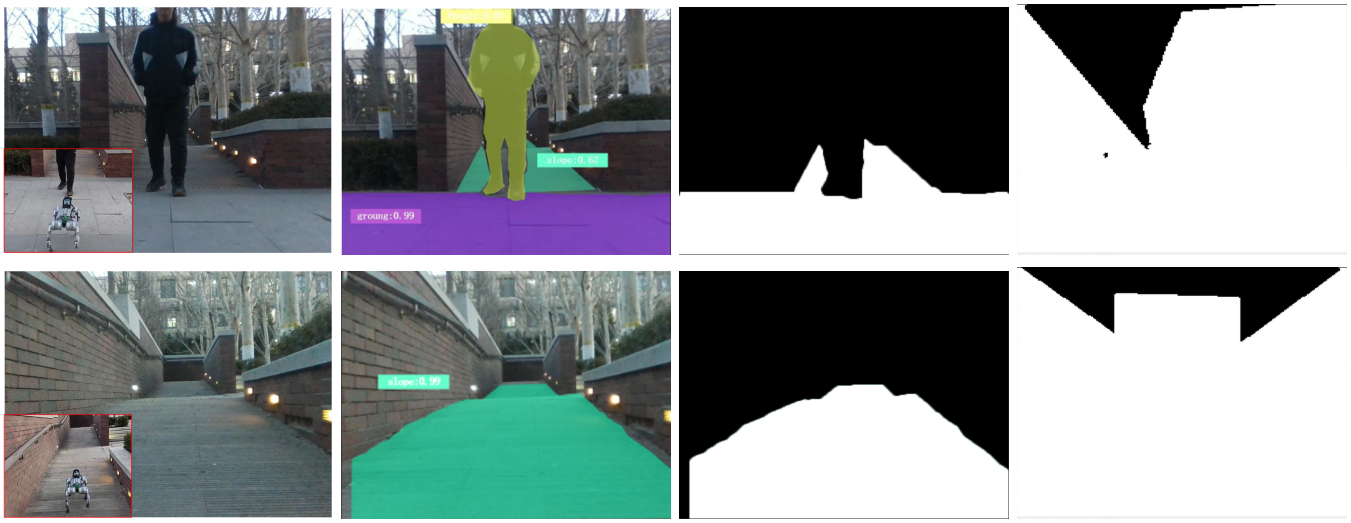
b) Training Details: To enhance semantic segmentation performance, image augmentation is applied, including rotation, cropping, and color look-up table (LUT) processing to simulate overexposure. Augmentation includes:

- Color LUT for data augmentation.
- Random rotation uniformly sampled from $[-5°, 5°]$
- Random crop uniformly sampled from [0.8, 1]

The test results of the improved semantic segmentation algorithm are shown in Fig. 2 (b).

c) Binarization: Since the robot's control strategy involves lifting its legs to a height smaller than the height of a step, stairs are considered static obstacles, and traversable terrains (flat ground, grass, gravel, slopes) are classified accordingly. Humans are treated as dynamic obstacles. The segmentation results are binarized, with traversable terrain as white (0) and non-traversable areas as black (255). The binarization result is shown in Fig. 2 (c). The binarization result is shown in Fig. 2 (c).

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(a) (b) (c) (d)

Fig. 2: Image processing results. (a) original image. (b) semantic segmentation result. (c) binarization result. (d) mapping result.

2) Contour Extraction: The road surface segmentation results are processed to focus only on traversable terrain contours. Contour extraction is performed on the binary image by checking surrounding white pixels. Internal contour points are turned black, while others remain part of the contour. The final result is a continuous contour curve, extracted using OpenCV.

3) Coordinate Mapping: The image recognition results in pixel coordinates need conversion to real-world coordinates for mapping onto the actual terrain, crucial for navigation. This paper establishes a pixel-to-ground conversion model using the camera's intrinsic and extrinsic parameters. However, due to perspective distortion when the camera is parallel to the ground, the conversion accounts for the camera's optical imaging principles and geometric relationships.

Given the camera intrinsic parameters:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 462.1 & 0 & 320.0 \\ 0 & 462.1 & 240.0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

The grid map dimensions 6×6,and the resolution is $0.05m$; The map center is used as an offset for adjustment:

$$x_{offset} = \frac{width}{2}, z_{offset} = height \quad (2)$$

For each pixel coordinate (μ, ν) in the image, the depth value d is obtained from the depth map.Using the camera's intrinsic matrix, the pixel coordinates (μ, ν) are converted into camera coordinates (x_c, y_c):

$$x_c = \frac{\mu - c_x}{f_x}, y_c = \frac{\nu - c_y}{f_y} \quad (3)$$

The camera coordinates are then transformed into real-world coordinates(X, Y, Z):

$$X = x_c Z, Y = y_c Z, Z = \frac{d}{\sqrt{x_c^2 + y_c^2 + 1}} \quad (4)$$

Finally, the transformed real-world coordinates(X, Y, Z) are mapped into the grid map coordinate system, where the grid map coordinates are (i, j):

$$i = \frac{x}{resolution} + x_{offset}, j = \frac{z}{resolution} + z_{offset} \quad (5)$$

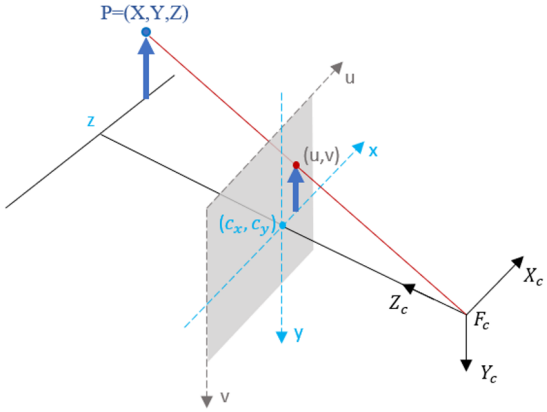


Fig. 3: Relationship Between Pixel Coordinates and Real-World Coordinates.

The coordinate transformation relationship is shown in the Fig. 3. The obstacle layer derived from the above transformation relationship is obtained, and the result is shown in Fig. 2 (d).

B. Semantic Costmap.

Continuing with the static layer, it is necessary to update the bounds and obtain the contour coordinate information of

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various terrains. Convert world coordinates into grid index coordinates for the current grid map. Use the grid index coordinates for cost value marking. The coordinate transformation formula is as follows:

Here, w_x and w_y represent the actual world coordinates of the terrain contour, while o_x and o_y denote the origin coordinates of the current map. Different types of terrain have varying impacts on the efficiency and safety of a quadruped robot's locomotion. The setting of cost value refers to the obstacle layer and expansion layer proposed by Lu et al. [22], setting the immovable obstacle grid as occupied (255), and setting the cost according to the safe walking degree of the quadruped robot, The cost functions corresponding to different terrains are shown in Table I.

TABLE I: Terrain Cost Value to Walking Strategy Table

Class	Cost	Linear (m/s)
ground	0	0.6
grass;slope	50	0.4
pebbles	100	0.2
steps;people	254	0.1

C. Inflation Layer

The inflation layer creates a buffer zone around areas marked as LETHAL_OBSTACLE, assigning progressively higher cost values to cells as they move away from these zones. This encourages the robot to avoid lethal obstacles.

D. Static Map Layer

The local semantic map, consisting of the inflation, static, and other layers, is used for local path planning, while the static and inflation layers make up the global map for global planning. For global path planning, the robot uses the global map to locate static obstacles. This paper uses depth information from a depth camera, converts it into LiDAR point clouds, and applies the GMapping algorithm to generate a high-precision 2D grid map as the static map layer.

III. Navigation System based on Local Semantic Map

The system consists of three modules: map construction, path planning, and motion control. Terrain semantic information is obtained through an RGB camera, geometric data through a depth camera, and robot pose through an IMU. Map construction provides both global and local maps, as discussed in Sec 2. The path planning module uses a search-based algorithm to find the optimal path, while the motion control module computes velocity based on the goal and walking strategy, sending commands to the higher-level control system. This enables end-to-end planning from input images to robot movement.

A. Path Planning

The system uses the move_base package, which provides configuration and execution interfaces for ROS navigation. The robot constructs a global costmap, and the global planner computes the optimal path using the A* algorithm. For local planning, a local semantic map and fused IMU/odometry data dynamically adjust the robot's trajectory using the Dynamic Window Approach (DWA). Terrain cost values are determined by the semantic cost layer, and the walking strategy generates optimal velocities.

Finally, control commands are executed by the robot's control system, enabling autonomous navigation. The process is detailed in Algorithm 1.

Algorithm 1 Navigation

Input: Global map G, Goal point $(x_{\text{goal}}, y_{\text{goal}})$, Semantic segmentation map S

Output: Motion commands: linear velocity v and angular velocity ω

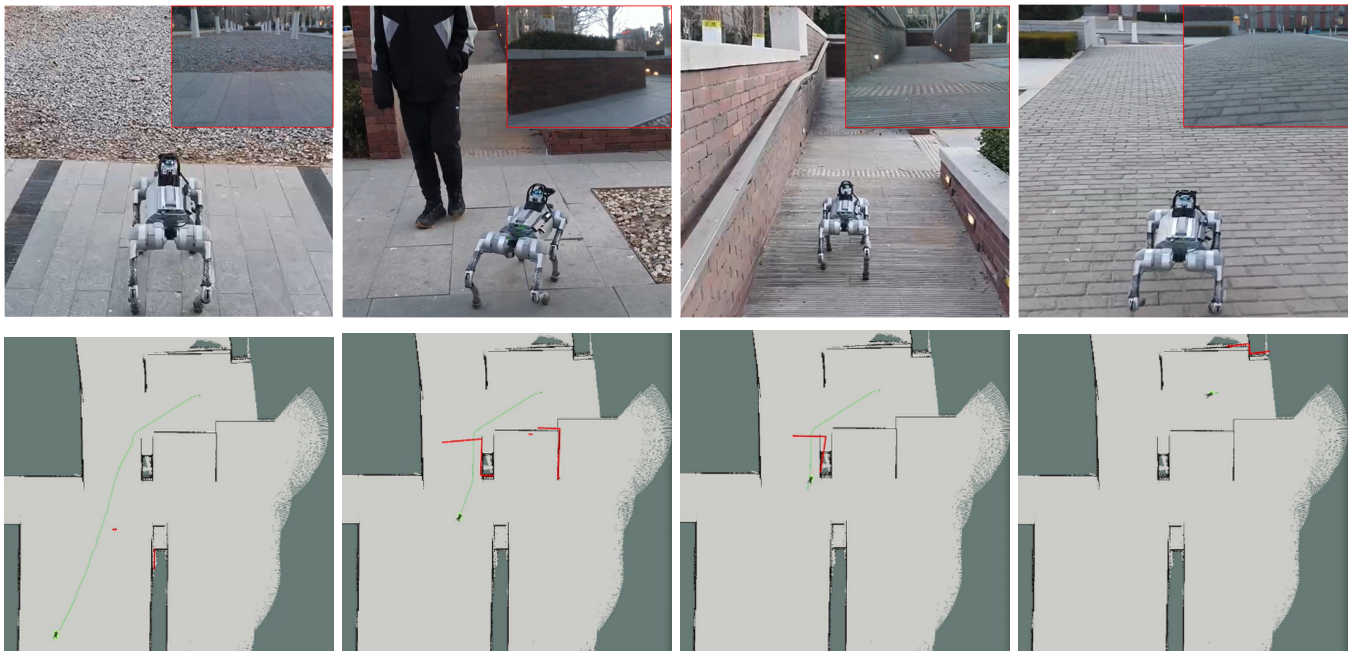
Input global map G
Input goal point $(x_{\text{goal}}, y_{\text{goal}})$
$path_global \leftarrow$ ASTAR$(G, (x_{\text{start}}, y_{\text{start}}), (x_{\text{goal}}, y_{\text{goal}}))$
while not reached goal **do**
    $S \leftarrow$ GetSemanticMap()
    $path_local \leftarrow$ DWA$(S, path_global)$
    if $path_local$ is None **then**
        Print "No valid local path found."
    else
        for each (p_x, p_y) in $path_local$ **do**
            $semantic_cost \leftarrow$ GetSemantic Cost(S, p_x, p_y)
            $v \leftarrow$ CalculateLinearVelocity $(semantic_cost)$
            $\omega \leftarrow$ CalculateAngularVelocity $(path_local, p_x, p_y)$
            SendMotionCommand(v, ω)
        end for
        $path_local \leftarrow$ UpdateGlobalPath$(G,$ robot_position, $(x_{\text{start}}, y_{\text{start}}), (x_{\text{goal}}, y_{\text{goal}}))$
    end if
end while

B. Motion Control

The motion control of a quadruped robot in complex environments requires high robustness to handle terrain variations and disturbances. This study uses deep reinforcement learning with the distributed proximal policy optimization (DPPO) algorithm for dynamic control tasks, enhanced by domain randomization for better transferability to real-world environments.

For safe and stable navigation, differentiated walking strategies are applied based on terrain types. The local target point coordinates are matched with the semantic cost layer of the local map to determine the terrain cost value, which then

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(a) (b) (c) (d)

Fig. 4: Experimental results of path planning for quadruped robots proposed in this paper. (The green line represents the global trajectory of the quadruped robot,and the black dot represents the actual position of the quadruped robot in the environment) (a) Give a goal. (b) Avoid dynamic obstacles. (c) Speed changes after terrain changes. (Ground to slope speed decreases.) (d) Arriving at the elevated terrain and reach the goal.

guides the appropriate linear velocity and walking strategy. The relationship between cost value and velocity is shown in TABLE I, with speed settings based on maximum safe speeds measured for various terrains to maintain stability.

IV. Expeirment

The method was validated using the Unitree Go2 quadruped robot with sensors including an Intel RealSense D435i camera and NVIDIA Jetson TX2 processor.

A. Semantic Segmentation

Six terrain types (e.g., flat ground, gravel, people, steps, slopes, and obstacles) are processed in this work. Due to vibrations during robot movement, overexposure of the RealSense D435i camera in sunlight, and image compression for network transmission, some terrains appear similar in low-quality images. Our method effectively defines terrain boundaries, even under challenging conditions such as overexposure or darkness. Data augmentation can improve classification accuracy in these cases. The semantic segmentation results for all terrains are shown in Fig. 3 (b).

TABLE II: Comparison Results for indoor and outdoor scenarios. IA represents the Image Augmentation.

Method	IA	FPS	Time	AP	AP_{75}
SparseInst [23]	-	35.6	28.1	79.0	87.1
SparseInst	✓	32.4	30.9	88.2	92.275

Average Precision (AP) is measured over multiple IoU's (Intersection of prediction and ground truth over Union of prediction and ground truth) thresholds, and mean Average Precision (mAP) is the average accuracy of multiple categories.The formula for the above evaluation indicators is as follows:

$$P = \frac{TP}{TP + FP} \tag{6}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{7}$$

$$AP = \frac{1}{N} \sum_{i=1}^{N} P_i \tag{8}$$

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i \tag{9}$$

In the formula, P represents accuracy,IOU is the intersection to union ratio, TP (True Positive) is the number of correctly classified positive samples, FP (False Positive) represents the number of negative samples incorrectly marked as positive. AP_{75} is average precision at IoU threshold of 0.75. The results evaluated on validation set are shown in Table II. There are no significant changes in real-time processing speed because the network structure is not modified. AP is 88.2%, indicating the overall high performance of the semantic segmentation system, especially given the low-quality images provided for

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prediction. Lower AP in much similar terrain types, a larger training dataset or higher quality images lead to higher AP.

B. Real-wolrd Experiments

To test the quadrupedal locomotion on a real-world robot, a 12-DOF robot is used. This paper conduct experiments in outdoor environments. The environment additionally includes a gravel path, slopes and people. The quadruped robot can avoid obstacles and walk naturally, as shown in Fig. 4. The robot can adapt to different environments by using the semantic cost layer proposed in this article to obtain the walking speed corresponding to the terrain. Finally, the robot successfully completed the inspection task in complex environments with elevated terrain and dynamic obstacles, and reached the goal. The success rate of the experiment can reach 85.7%.

V. Conclusion

This paper proposes a quadruped robot navigation method based on a local semantic map, enabling efficient operation on elevated terrains while ensuring safe and robust navigation during inspections. The approach integrates both global and local terrain representations, trajectory generation, and adaptive walking strategies. It effectively addresses navigation challenges in complex environments with dynamic obstacles, making it well-suited for inspection tasks. For further research, we will combine quadruped robots with multiple robots to conduct inspections and ensure environmental safety.

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